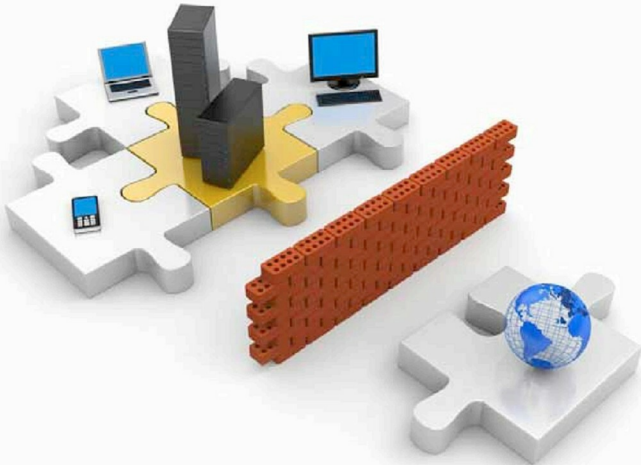


Build Your Own Firewall Using IPTables

A firewall monitors the incoming and outgoing traffic flow in the computer network such that every packet that enters or leaves the network has to pass through it. Based on the header information, a set of policies is verified and suitable action is taken on the packet, with the firewall either allowing authorised traffic to enter/leave the network or blocking unauthorised packets.



IPTables is the firewall used on the Linux platform. Prior to IPTables, IPfwadm and IPChains were among the most popular Linux firewalls. They had certain imperfections, which were fixed, resulting in a new product from the Netfilter organisation called IPTables.

IPTables is a faster and more secure alternative to IPChains and IPfwadm. RedHat and Fedora Linux have made IPTables their default pre-installed firewall package.

Packet processing in IPTables

Every packet passes through a series of built-in queues called *Tables* for processing. Each table is associated with a certain packet activity and is handled by a filtering chain where firewall policies can be placed.

Basically, there are four tables.

- Filter table: This is associated with packet filtering.
- NAT (Network Address Translation) table: This is responsible for alternating the source and/or destination

IP address (applicable in case more than one network interface is enabled).

- Mangle table: This is a combination of a filter and NAT table, and it is related to the modification of the Quality of Service bits in the TCP header.
- Raw table: This is associated with those packets that are not traced.

The Filter table is the default table and has three in-built chains.

- The 'Input' chain controls traffic entering/destined for the system.
- The 'Forward' chain controls traffic travelling/being routed through the system.
- The 'Output' chain controls traffic leaving or originating from the system.
- The NAT table has three in-built chains. The 'Prerouting' chain is used to NAT packets after changing their destination address.